

COLLABORATIVE DISCUSSION 1: PEER RESPONSES

Ariel Mella

12701097

Unit 3

Machine Learning

University of Essex Online

CONTENTS

MY PEER RESPONSES TO OTHERS.....	3
Peer Response: To Peer A on HSE Ireland ransomware (2026).....	3
Peer Response: To Peer B on pharmaceutical Industry 4.0 (2026).....	5
PEER RESPONSES TO ME.....	7
Peer Response: From Peer Response 1 (2026).....	7
Peer Response: From Peer Response 2 (2026).....	10

MY PEER RESPONSES TO OTHERS

Peer Response: To Peer A on HSE Ireland ransomware (2026)

Submitted Wednesday 13 May 2026 at 10:34 UK time, in response to a peer's initial post on the 2021 ransomware attack on Ireland's Health Service Executive and Industry 4.0 in healthcare.

The authorities reported in 2022 that HSE had weak IT infrastructure, over-reliance on legacy systems, a flat network, and no standardised backup and recovery approach across data centres (Office of the Controller and Auditor General, 2022). That suggests preventive measures should have included basics: network segmentation, monitoring, legacy-system replacement when possible and more importantly backups that are tested with a comprehensive recovery process for the critical systems that must operate when incidents occur.

In terms of recovery, this is where Industry 5.0's resilience pillar becomes practical (European Commission, 2021). I believe there is no such thing as total security, but only degrees of it, so healthcare organisations should plan not only to prevent and respond, but also to restore critical services in a production-like, tested way. Moore et al. (2023) show that staff had to rely on paper-based workarounds, but these created safety and workload problems while systems remained down. This means "manual alternatives" are necessary, and often recommended within continuity practice, but not enough unless designed, trained and tested as part of the operating model.

NIST (2024) frames Recover as a core cybersecurity function, which supports your point about resilience. My question is: should healthcare digital-transformation programmes be approved only when they include appropriate recovery testing and safe manual fallback plans, not just efficiency and AI benefits?

References

European Commission (2021) Industry 5.0: towards a sustainable, human-centric and resilient European industry. Luxembourg: Publications Office of the European Union. Available at:

<https://op.europa.eu/en/publication-detail/-/publication/468a892a-5097-11eb-b59f-01aa75ed71a1/language-en> (Accessed: 13 May 2026).

Moore, A., Yousef, T., Garde, S. and Coiera, E. (2023) 'Impact of the 2021 cyber-attack on Ireland's Health Service Executive: a qualitative study of staff experiences', *BMJ Health & Care Informatics*, 30(1), e100792. Available at: <https://doi.org/10.1136/bmjhci-2023-100792> (Accessed: 13 May 2026).

NIST (2024) The NIST Cybersecurity Framework (CSF) 2.0. NIST Cybersecurity White Paper, NIST CSWP 29. Gaithersburg, MD: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.CSWP.29> (Accessed: 13 May 2026).

Office of the Controller and Auditor General (2022) Report on the 2021 cyber-attack on the Health Service Executive. Dublin: Office of the Controller and Auditor General. Available at: <https://www.audit.gov.ie/en/find-report/publications/> (Accessed: 13 May 2026).

Peer Response: To Peer B on pharmaceutical Industry 4.0 (2026)

Submitted Wednesday 13 May 2026 at 10:01 UK time, in response to a peer's initial post on Industry 4.0 in the pharmaceutical sector (vaccine development, Miltenyi Biotec ransomware, EMA breach).

Hi, I agree with the challenge: cybersecurity within Industry 4.0 is often underestimated, and the focus remains on gains. Evidence shows investment in protecting those gains lags behind; Accenture (2025) reports that only 28% of organisations include cybersecurity in transformation initiatives, and the World Economic Forum (2025) finds that less than half of CEOs believe their organisations invest enough.

In terms of preventive measures, they already exist; the challenge is finding the right investment-reward equation. For the ransomware example, a Zero Trust Architecture could have helped by requiring continuous verification and least-privilege access, reducing privilege escalation or lateral movement across interconnected pharmaceutical systems (Rose et al., 2020). Automated anomaly detection and network segmentation would support earlier detection before operational systems are compromised.

The second point is recovery. NIST (2024) frames cybersecurity through Govern, Identify, Protect, Detect, Respond and Recover. All are equally important because there is no such thing as total security, only degrees of security. Therefore, organisations should plan not only to prevent and respond, but also to restore systems from backups, prioritise critical services, and test recovery procedures in production-like environments before incidents occur (McBride et al., 2020).

Merck's NotPetya incident disrupted manufacturing, research and sales, while West Pharmaceutical Services' 2026 cyberattack encrypted systems and disrupted global operations (Crosignani, Macchiavelli and Silva, 2020; West Pharmaceutical Services, 2026).

Therefore, if Industry 5.0 frames resilience as a core value, what still needs to be addressed so funding and efforts improve resilience effectively: regulation, accountability, mandatory testing, or something else?

References

- Accenture (2025) State of cybersecurity resilience 2025. Dublin: Accenture. Available at: <https://www.accenture.com/us-en/insights/security/state-cybersecurity> (Accessed: 13 May 2026).
- Crosignani, M., Macchiavelli, M. and Silva, A.F. (2020) Pirates without borders: the propagation of cyberattacks through firms' supply chains. Federal Reserve Bank of New York Staff Report No. 937. Available at: https://www.newyorkfed.org/research/staff_reports/sr937 (Accessed: 13 May 2026).
- McBride, T., Ekstrom, M., Lusty, L., Sexton, J. and Townsend, A. (2020) Data integrity: recovering from ransomware and other destructive events. NIST Special Publication 1800-11. Gaithersburg, MD: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.SP.1800-11> (Accessed: 13 May 2026).
- NIST (2024) The NIST Cybersecurity Framework (CSF) 2.0. NIST Cybersecurity White Paper, NIST CSWP 29. Gaithersburg, MD: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.CSWP.29> (Accessed: 13 May 2026).
- Rose, S., Borchert, O., Mitchell, S. and Connelly, S. (2020) Zero trust architecture. NIST Special Publication 800-207. Gaithersburg, MD: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.SP.800-207> (Accessed: 13 May 2026).
- West Pharmaceutical Services (2026) 'Cybersecurity incident disclosure', West Pharmaceutical Services Investor Relations, 12 February. Available at: <https://investor.westpharma.com/> (Accessed: 13 May 2026).
- World Economic Forum (2025a) Global cybersecurity outlook 2025. Geneva: World Economic Forum. Available at: <https://www.weforum.org/publications/global-cybersecurity-outlook-2025/> (Accessed: 13 May 2026).

PEER RESPONSES TO ME

Peer Response: From Peer Response 1 (2026)

Received Thursday 7 May 2026 at 9:51 UK time, in response to my initial post on Industry 4.0 and 5.0 in the automotive sector. Reproduced verbatim with the author's references.

Your post makes a convincing link between AI-enabled automotive production and the risks created by highly connected Industry 4.0 systems. BMW's predictive maintenance example is interesting because it shows AI being used proactively: production data helps identify early fault patterns and guide maintenance responses before disruption escalates (BMW Group, 2023). However, this proactive logic should also apply to cyber and operational resilience, not only mechanical maintenance.

The Jaguar Land Rover incident strengthens this point. Reuters (2025) reported that Jaguar Land Rover's manufacturing operations returned to normal after a six-week cyber shutdown, with disrupted supply chains and related costs reaching £196 million, demonstrating that "smart factory" performance also depends on cyber resilience, business continuity and supply-chain stability. From an Industry 5.0 perspective, resilience must be designed into the system before disruption occurs. Metcalf (2024) argues that Industry 5.0 rebalances the technology-centred logic of Industry 4.0 around human-centricity, sustainability and resilience. Similarly, van Erp et al. (2024) describe Industry 5.0 as integrating these principles into industrial value creation, suggesting that automotive firms need socio-technical safeguards as much as technical innovation.

One prevention measure is treating cyber risk as part of production governance. NIST's Cybersecurity Framework 2.0 is relevant here because it frames cyber resilience around governance, identification, protection, detection, response and recovery, including supply-chain risk (NIST, 2024). For automotive manufacturers, this could involve stricter supplier access controls, compartmentalised operational technology networks, rehearsed shutdown/recovery plans and anomaly monitoring across IT and factory systems.

There is also a human-centred point. AI may improve productivity, but the OECD notes risks around privacy, work intensity, bias and changing skills requirements (OECD, 2023). Therefore, Industry 5.0 should build on the automation gains achieved through Industry 4.0 by using AI to support workers, improve decision-making and strengthen organisational resilience.

References

BMW Group (2023) 'Smart maintenance using artificial intelligence', BMW Group PressClub Global, 27 November. Available at: <https://www.press.bmwgroup.com/global/article/detail/T0438145EN/smart-maintenance-using-artificial-intelligence?language=en> (Accessed: 7 May 2026).

Metcalf, G.S. (2024) 'An introduction to Industry 5.0: history, foundations, and futures', in Nousala, S., Metcalf, G.S. and Ing, D. (eds) *Industry 4.0 to Industry 5.0: explorations in the transition from a techno-economic to a socio-technical future*. Singapore: Springer, pp. 1-29. Available at: https://doi.org/10.1007/978-981-99-9730-5_1 (Accessed: 7 May 2026).

NIST (2024) *The NIST Cybersecurity Framework (CSF) 2.0*. NIST Cybersecurity White Paper, NIST CSWP 29. Gaithersburg, MD: National Institute of Standards and Technology. Available at: <https://doi.org/10.6028/NIST.CSWP.29> (Accessed: 7 May 2026).

OECD (2023) *OECD Employment Outlook 2023: artificial intelligence and the labour market*. Paris: OECD Publishing. Available at: <https://doi.org/10.1787/08785bba-en> (Accessed: 7 May 2026).

Reuters (2025) 'Jaguar Land Rover's UK production returns to normal after weeks-long cyber shutdown', Reuters, 14 November. Available at: <https://www.reuters.com/business/retail-consumer/jaguar-land-rovers-uk-production-returns-normal-after-weeks-long-cyber-shutdown-2025-11-14/> (Accessed: 7 May 2026).

van Erp, T., Carvalho, N.G.P., Gerolamo, M.C., Gonçalves, R., Rytter, N.G.M. and Gladysz, B. (2024) 'Industry 5.0: A new strategy framework for sustainability management and beyond', *Journal of Cleaner Production*, 461, article 142271. Available at: <https://doi.org/10.1016/j.jclepro.2024.142271> (Accessed: 7 May 2026).

Peer Response: From Peer Response 2 (2026)

Received Monday 11 May 2026 at 2:09 UK time, in response to my initial post on Industry 4.0 and 5.0 in the automotive sector. Reproduced verbatim with the author's references.

Dear Ariel,

I really enjoyed reading your post. It effectively connects real-world industrial applications with broader Industry 4.0 concepts such as smart factories, predictive maintenance, and quality optimisation through the experience of a leading automotive manufacturer.

I particularly agree with your emphasis on predictive maintenance, which has become a central pillar of Industry 4.0 due to its ability to reduce unexpected downtime, improve equipment availability, and support data-driven decision-making within manufacturing systems (Zonta et al., 2020).

Your discussion of enhanced product quality through Quality 4.0 is also highly relevant in the automotive sector, where AI-driven quality control contributes to vehicle reliability, organisational reputation, customer loyalty, and reduced warranty costs (Zhou et al., 2019).

Your example of the Jaguar Land Rover cybersecurity incident clearly highlights the broad impact such events can have on Industry 4.0 organisations. Although the affected stage of the manufacturing process was not specified, the disruption may have originated from a supply chain vulnerability, given the critical role of interconnected suppliers and third-party systems in Industry 4.0 environments (Zarreh et al., 2019).

To reduce the likelihood of similar incidents, manufacturing leaders recommend evidence-based cybersecurity approaches, including continuous risk assessments and adherence to industry standards (Okomanyi, Sherwood and Shittu, 2024). Among these, the National Institute of Standards and Technology Cybersecurity Framework (NIST CSF) is recognised as one of the leading frameworks for cybersecurity assurance in the automotive industry (Burkacky et al., 2020).

In addition, specific frameworks have been proposed to address supply chain security concerns within Industry 4.0 environments. Sowan, Zighan and Altarawneh (2025) proposed a multi-layered framework that integrates technologies such as blockchain, AI, IoT, and big data analytics to strengthen security through decentralised data management, real-time monitoring, and automated threat detection. Their framework also combines technological solutions with organisational strategies, including risk assessment, employee training, and collaborative risk management.

To conclude, several approaches can be adopted to strengthen cybersecurity assurance. Most industry leaders recommend well-established frameworks such as the NIST Cybersecurity Framework (NIST CSF), while emerging supply chain security challenges may require newer frameworks grounded in Industry 4.0 technologies.

References

- Burkacky, O., Deichmann, J., Klein, B., Pototzky, K. and Scherf, G. (2020) Cybersecurity in automotive: mastering the challenge. McKinsey & Company, June. Available at: <https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/cybersecurity-in-automotive-mastering-the-challenge> (Accessed: 11 May 2026).
- Okomanyi, A.O., Sherwood, A.R. and Shittu, E. (2024) 'Exploring effective strategies against cyberattacks: the case of the automotive industry', *Environment Systems and Decisions*, 44(4), pp. 779-809. Available at: <https://doi.org/10.1007/s10669-024-09971-0> (Accessed: 11 May 2026).
- Sowan, B.I., Zighan, S. and Altarawneh, A. (2025) 'Supply Chain 4.0: driving operational efficiency, resilience, and cybersecurity using advanced technologies', 2025 1st International Conference on Computational Intelligence Approaches and Applications (ICCIAA), pp. 1-8. Available at: <https://doi.org/10.1109/ICCIAA65327.2025.11013378> (Accessed: 11 May 2026).
- Zarreh, A., Wan, H., Lee, Y., Saygin, C. and Janahi, R. (2019) 'Cybersecurity concerns for total productive maintenance in smart manufacturing systems',

Procedia Manufacturing, 38, pp. 532-539. Available at:
<https://doi.org/10.1016/j.promfg.2020.01.067> (Accessed: 11 May 2026).

Zhou, F., Lin, Y., Wang, X., Zhou, L. and He, Y. (2019) 'Supplier portfolio of key outsourcing parts selection using a two-stage decision making framework for Chinese domestic auto-maker', Computers & Industrial Engineering, 128, pp. 559-575. Available at: <https://doi.org/10.1016/j.cie.2018.12.014> (Accessed: 11 May 2026).

Zonta, T., da Costa, C.A., da Rosa Righi, R., de Lima, M.J., da Trindade, E.S. and Li, G.P. (2020) 'Predictive maintenance in the Industry 4.0: a systematic literature review', Computers & Industrial Engineering, 150, article 106889. Available at: <https://doi.org/10.1016/j.cie.2020.106889> (Accessed: 11 May 2026).